

qa_crib.md — 10 likely judge questions, 2-line answers (C6)

Longer rehearsed versions: `SPEAKER_SCRIPT.md` §Q&A. Universal fallback: "Every number on our slides carries a ledger row with the producing command — I can pull it up." Always true; buys ten seconds.

1. Your headline says $n=7$ but the hardware ran $n=6$ — why? $n=7$ is the last point of the compilation trend [R046]. The hardware job carries a deliberately-dead 8,850-gate control that dominates cost — $\sim 4\times$ more at $n=7$ for zero added contrast; the full arithmetic is appendix A7 [R055].

2. How can compilation "succeed" at $n=7$ if hardware "fails harder" there? The trend is a ratio (exact $\sim 4^{(n+1)}$ over AQC $\sim$ linear); hardware survival decays with the AQC circuit's absolute depth, which still grows. Measured effective error is $\sim 6e-3$ /gate; visibility needs $\sim 2e-3$ — one device generation, not a protocol change [R054, R055].

3. Isn't the Loschmidt echo simply better? For the scalar overlap, yes — $4\text{--}6\times$ cheaper and $27\times$ more accurate on hardware, confirmed on an independent run over a $2\times$ -extended time window; we say so on the slide [R043, R044, R059]. It returns one magnitude; only our arm returns the phase and the per-observable breakdown [R042].

4. Do you actually know why the AQC arm failed on hardware? Partially, and we say which part. Gate error plus ancilla dephasing predicts 0.186; measured 0.026 — a $\sim 7\times$ residual we label unattributed (candidates: layout edges, crosstalk, coherent error) rather than invent a cause [R054].

5. Why believe the $36\times$ at all, after the hardware negative? They are different claim types: $36\times$ is a transpiled gate count, deterministic and reproducible from the repo [R046, R052]; the negative is about today's error rates, and we pre-registered both predictions so neither claim can quietly absorb the other [R054].

6. The compressed circuit's process infidelity is 0.96 — isn't that disqualifying? It would be anywhere else. A Hadamard test's controlled gate acts on exactly one prepared state, and AQC is compressed against that state; we state on the slide that reusing it on any other input is illegitimate [R046, R051].

7. Can the $\langle Q \rangle$ error bar be used to correct the data? No — diagnostic only. Post-selecting on it biases the shadow estimator, whose unbiasedness rests on uniform random bases; that failure class is documented in our bug log (B04) [R044, R036].

8. Did you try SKQD / sample-based diagonalisation? Yes — and mapped where it works: it needs a localised ground state. Our benchmark is hopping-dominated ($\Delta/J=0.38$, needs 86.6% of the sector); at $\Delta/J=5$ it needs 2.7%, and the same boundary reappears on 2D Fermi-Hubbard in U/t . Configuration recovery did earn its keep on real shots [R047, R050, R037].

9. Is this really the first open implementation of the PRL? We searched and found none, with the scope recorded and date-stamped in the repo; we would not defend priority if challenged — the characterisation (costs, failure boundaries, hardware) is ours either way [PRIOR_ART_SEARCH.md, SRCH].

10. Do you claim any speedup over classical methods? No. The benchmark is $2\text{--}7$ qubits, diagonalisable on a laptop — deliberately, so every estimate is graded against exact truth; our claims are about measurement protocols and circuit costs, not classical intractability.

11. Have you compared your method against pure AQC — no Hadamard test, no shadow? Yes, and it's a clean result: pure AQC (compress the state, measure ordinary observables directly, no ancilla at all) is *safe* — worst error 0.014–0.018 on $\langle Q \rangle / \langle H \rangle / \langle Z_0 \rangle$ at $n=3/5/7$, because those expectation values are exactly phase-invariant (proven algebraically and confirmed by an explicit phase-perturbation test) [R060]. The phase trap only bites once

AQC is embedded in an interferometric circuit like ours, where the SAME compiled circuit's own state-fidelity certificate ($1.4\text{e-}4$ – $3.0\text{e-}4$, looks perfect) hides a χ error of 1.22–1.35 on a quantity bounded by 1 [R046]. So AQC itself isn't broken — the hazard is specific to interferometric verification, which is exactly what our method exists to catch.